

Non-Functional Requirements

CodeClash

Performance

- The system shall load the initial application shell within 3 seconds on a standard broadband connection
- The matchmaking system shall pair users within 30 seconds under normal load conditions
- Code execution results shall be returned to the user within 10 seconds for standard submissions
- The system shall support at least 100 concurrent active matches without degradation in response time

Scalability

- The matchmaking service shall be designed to scale horizontally to accommodate a growing user base
- The code execution service shall be decoupled from the primary backend to allow independent scaling under high submission load
- The database shall be designed to handle a growing match history and user base without requiring structural changes

Security

- All user passwords shall be stored as salted hashes using the latest industry standard on Cognito.
- All data in transit shall be encrypted using TLS (https:// instead of http).
- JWT tokens shall expire after a defined period and require refresh
- User-submitted code shall never be executed outside of an isolated sandboxed environment
- The execution sandbox shall enforce strict CPU time caps, memory quotas, (and string cap?) and network access disablement
- All API endpoints shall enforce authentication and authorisation checks before processing requests
- The system shall implement rate limiting on authentication endpoints to prevent brute force attacks

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Reliability

- The system shall handle WebSocket disconnections gracefully, preserving match state for a reconnecting user for at least 60 seconds

- A failure in the code execution service shall not crash or affect the primary backend server
- The system shall have an uptime of at least 99% during active usage periods
- The system shall log all errors and exceptions to enable debugging and monitoring

Usability

- The application shall be fully functional on the latest versions of Chrome, Firefox, Safari, and Edge
- The interface shall provide immediate visual feedback for all user actions within 200ms
- Error messages shall be clear, specific, and actionable rather than generic
- The code editor shall provide syntax highlighting and auto-indentation for all supported languages
- The application shall be responsive across desktop screen sizes, with the programming battle interface optimised primarily for desktop

Maintainability

- The codebase shall follow consistent coding standards enforced by ESLint
- All core business logic shall have a minimum of 80% unit test coverage
- The system shall follow a modular architecture, ensuring that changes to one service do not require changes to others
- All environment-specific configuration shall be managed through environment variables and never hardcoded

Testability

- All core services shall have unit tests written in Jest
- Critical user flows shall be covered by integration tests
- The code execution service shall have test cases covering edge cases including infinite loops, memory overflows, and malformed or malicious input
- The system shall support mock implementations of the Judge0 service for testing purposes without requiring a live execution environment

Deployability

- The entire system shall be deployable using a single docker-compose up command in a development environment
- The backend and code execution engine shall be containerised using Docker
- The system shall be deployed to a publicly accessible cloud environment before each demo
- The main branch shall always reflect a working deployable version of the system

Auditability

- The system shall maintain logs of all match outcomes, Elo changes, and user reports
- All anti-cheat flags shall be logged with timestamps and associated user identifiers for review